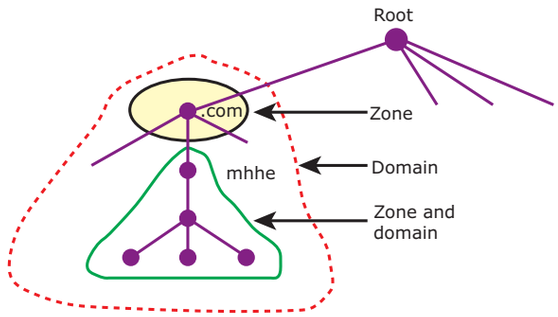


Figure 12.9 Zone and Domains

12.5.4 Resolver

Domain Name System is a client/ server application. A host system need to map domain name to IP address or vice versa according to the call and that work is done by resolver. Resolver either asks server to provide information about IP address. If it doesn't find any information, then it sends the request to other servers and so on. Once the resolver receives the mapping, it checks whether it is an error or resolution (mapping) and provides result according to that.



The resolver is a program which is responsible for initiating the translation of a domain name into an IP address. Since a resolver is stored in the host, There is no need of any protocol to form a connection between the resolver and the user program.

12.5.5 How DNS works?

When the user enters the URL (consists of protocol, domain name, folder name, file name) in the browser, the system first checks its DNS cache for the corresponding IP address. If the IP address is found in the cache then the information is retrieved from cache. If not, then the system needs to perform DNS query i.e., the system needs to query the resolver about the IP address from Internet Service Provider (ISP). Each resolver has its own cache and if it is found in that then that information is retrieved. If not, then the query is passed to next domain

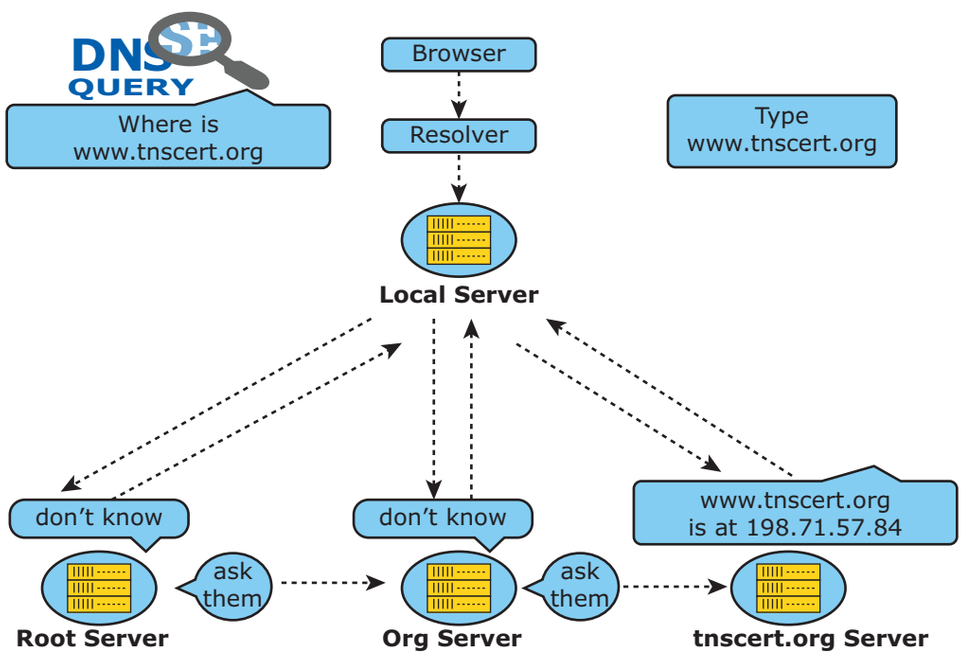


Figure 12.10 workflow of basic DNS

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